



Computer Vision

PECAIML701B

Topic - Thresholding and Binarization

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Stream :- AIML

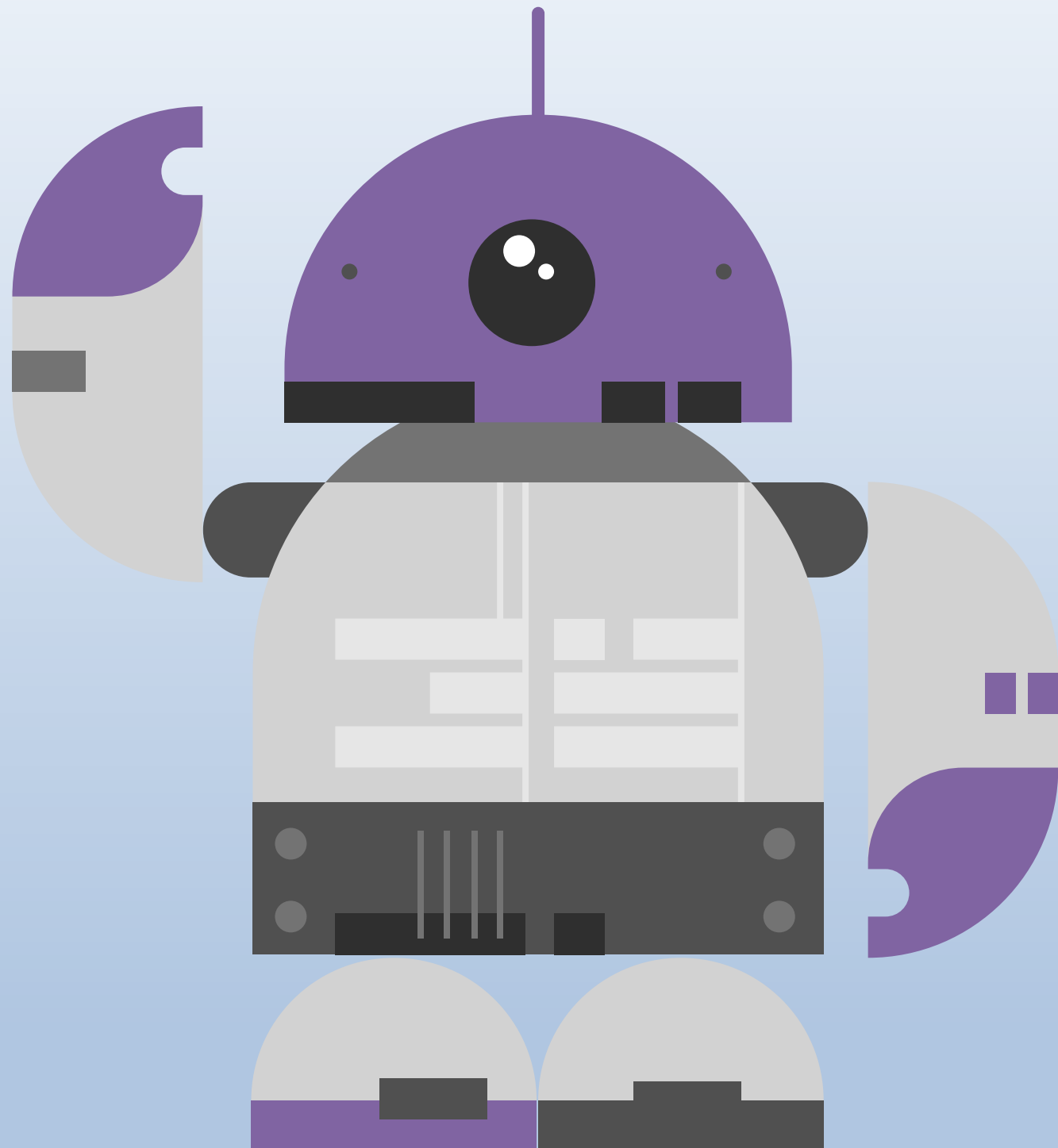
Class Roll :- 31

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Introduction to Thresholding

Definition: Thresholding is a process of converting a grayscale image into a binary image by comparing pixel intensities against a threshold value.

Binary Image: A binary image contains only two possible pixel values — usually **0 (black)** for background and **255 (white)** for foreground.

Key Idea: If the pixel value is greater than or equal to the threshold, it becomes white; otherwise, it becomes black.

Importance: Helps in isolating regions of interest and reducing computational complexity.

Purpose of Thresholding



Simplifies Image Data: Reduces multi-level intensity data to just two levels, making analysis easier.

Separates Objects from Background: Useful for identifying shapes, text, or features in an image.

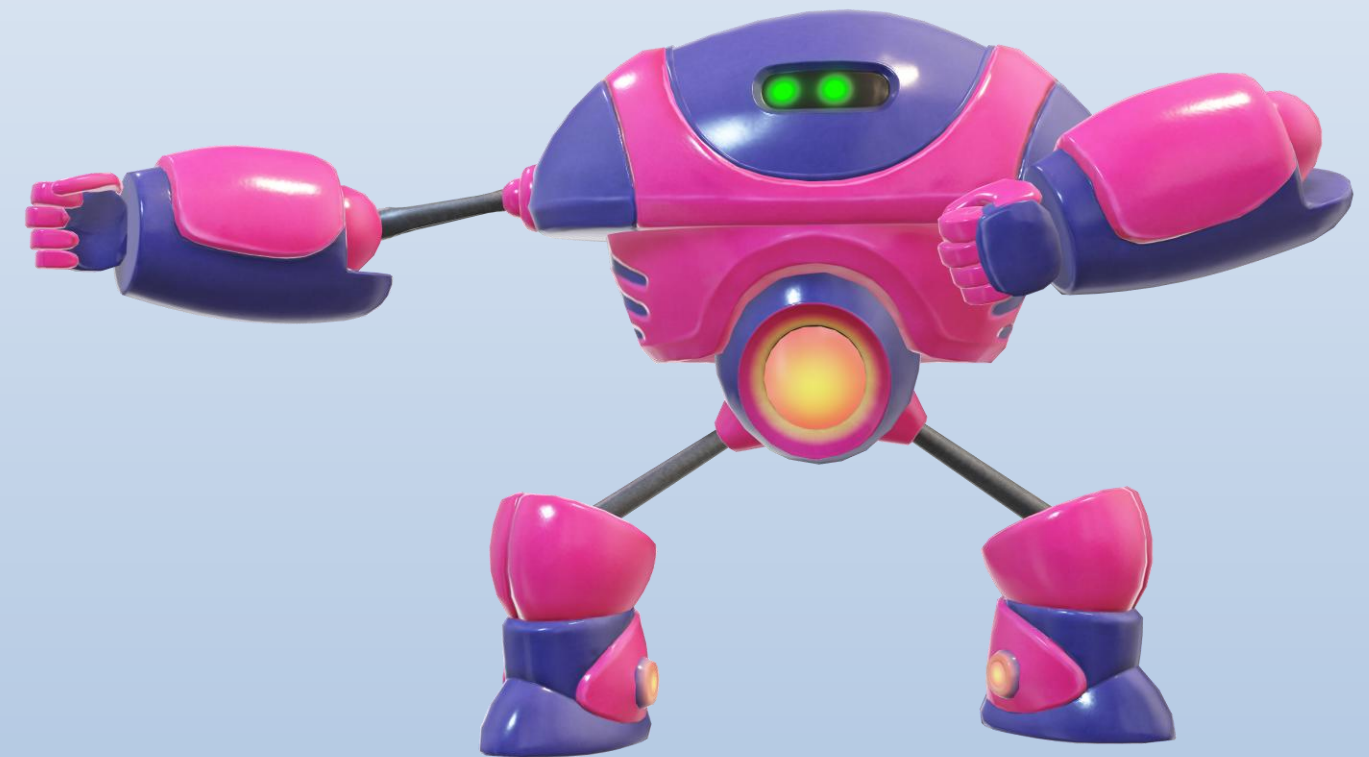
Preprocessing Step: Often used before contour detection, shape recognition, OCR, and segmentation.

Applications:

- Medical imaging (e.g., detecting tumors)

- Industrial inspection (e.g., defect detection)

- Document scanning (e.g., removing noise and enhancing text)



Types of Thresholding

There are three common thresholding techniques:

1. **Global Thresholding:** Uses a single threshold for the entire image.
2. **Adaptive Thresholding:** Uses different thresholds for different image regions.
3. **Otsu's Method:** Automatically determines the optimal threshold based on the image histogram.



Global Thresholding



- **Process:**

- Choose a threshold value T manually.
- Compare each pixel's intensity to T and assign black or white accordingly.

- **Advantages:**

- Simple and fast to implement.
- Works well with uniform lighting conditions.

- **Disadvantages:**

- Fails when there is non-uniform illumination or varying shadows.

- **Example:**

For $T = 127$, all pixels > 127 become white, others become black.

Adaptive Thresholding



Concept: Instead of a single global threshold, different parts of the image have their own local threshold.

Process:

- Divide the image into small regions (blocks).

- Calculate a threshold for each block (e.g., using mean or Gaussian-weighted sum of neighborhood values).

Advantages:

- Handles uneven lighting, shadows, and gradients effectively.

Disadvantages:

- More computationally expensive than global thresholding.

Example Use Case:

- Digitizing old documents with faded text and dark spots.

Otsu's Method

Concept : Finds an optimal threshold automatically without user input.

How it Works:

- Analyzes the image histogram.

- Selects a threshold that minimizes intra-class variance (variation within foreground and background) or maximizes inter-class variance.

Advantages:

- Fully automatic and objective.

- Works well for bimodal histograms (two distinct peaks).

Limitations:

- Less effective for images without clear intensity separation.



Binary Image Examples

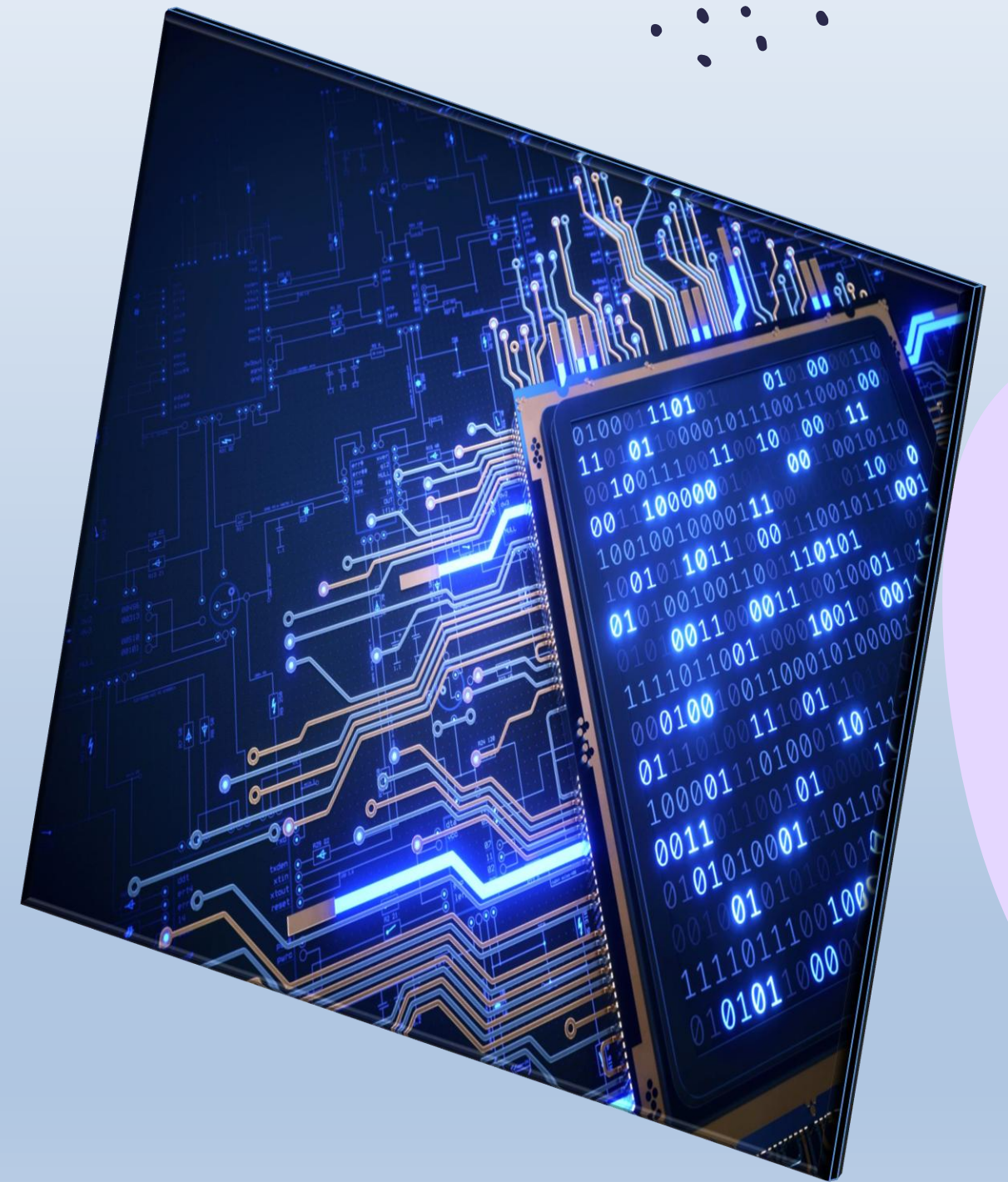
Examples should show:

Original Image: A grayscale picture of a handwritten note.

Global Threshold: Clear black-and-white image if lighting is uniform.

Adaptive Threshold: Better results when parts of the note are in shadow.

Otsu's Method: Automatically chosen threshold for clear separation of ink and paper.



When to Use Which Method

Global Thresholding :

Use when lighting is uniform and objects have distinct intensity differences from the background.

Adaptive Thresholding:

Use when lighting varies across the image or when shadows exist.

Otsu's Method:

Use when you don't know the threshold in advance and the histogram has two clear peaks.



CONCLUSION

Thresholding is a **crucial preprocessing technique** in image processing.

The choice of method depends on **lighting conditions, image complexity, and automation needs**.

Combining thresholding with other preprocessing steps can greatly improve object detection, recognition, and analysis results.

Key takeaway: Choose the simplest method that produces accurate results for your image conditions.



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CV_CA1 Quiz

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Your email (rajeshraftar.aiml2022@nsec.ac.in) was recorded when you submitted this form.

Name: *

Rajesh Raftar

University Roll No.: *

10930622031

In computer vision, what does object detection aim to do?

*



Convert images into audio

- ☐ Classify the entire image
- ☒ Locate and identify objects within an image
- ☐ Reduce the image resolution

Which domain is most suitable for applying Fourier Transform in image processing?

*

- ☐ Spatial domain
- ☒ Frequency domain
- ☐ Temporal domain
- ☐ RGB domain

What is the output of a thresholding operation in image processing?

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- ☐ A grayscale image
- ☐ A color-enhanced image
- ☒ A binary image
- ☐ A segmented image with 256 levels

What is the purpose of the convolution operation in image processing?

*

- ☐ To compress image size
- ☒ To detect patterns or features in images
- ☐ To convert color images to grayscale

- ☐ To normalize pixel values

Which of the following operations is used to remove noise from an image?

*

- ☐ Histogram equalization
- ☐ Convolution with a high-pass filter
- ☒ Gaussian smoothing
- ☐ Edge Detection

What does a pixel represent in a digital image? *

- ☐ A compressed block of colors
- ☐ A graphical object
- ☒ The smallest unit of an image
- ☐ A 3D coordinate

In computer vision, which technique is often used to correct geometric distortions in images?

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- ☐ Color segmentation
- ☐ Histogram equalization
- ☒ Image warping
- ☐ K-means clustering

Which of the following is a color model commonly used in computer vision?

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- ☐ PDF
- ☒ RGB
- ☐ XML
- ☐ JPG

What does image sampling primarily refer to in digital image processing?

*

- ☐ Converting a color image to grayscale
- ☒ Measuring the intensity of pixels at discrete points in the image
- ☐ Assigning discrete gray levels to pixel intensities
- ☐ Removing noise from the image

In image quantization, increasing the number of quantization levels will generally:

*

- ☐ Increase image distortion
- ☐ Reduce the image file size significantly
- ☒ Increase the image's visual quality
- ☐ Reduce the spatial resolution

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